

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages	Python, C++, TypeScript, JavaScript, SQL, Java, Bash, HTML/CSS
Backend & APIs	Node.js, Express, FastAPI, Fastify, REST APIs, async services, integration testing, design patterns, authentication, rate limiting, audit logging
Systems & Performance	Linux perf profiling, cache-aware design, OpenMP, AVX intrinsics, SIMD, from-scratch autodiff, Apache Spark / PySpark, Databricks, Spark UI
Cloud & DevOps	AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux
Databases & Web	MongoDB, PostgreSQL, Redis, SQLite, React, Electron, Webpack

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Engineered **multi-tenant NHS eTriage backend** (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions** as a production-grade system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines; contributed to **\$340K+** in revenue.
- Owned end-to-end code quality and maintainability across the eTriage stack: code reviews, integration tests, production-log monitoring, fixes for tenant-specific incidents, and security hardening including authentication, rate limiting, and remediation of **20+ VAPT findings**.
- Architected a full-stack data annotation platform (React, Node.js, MongoDB) used daily by the AI team; replaced spreadsheet workflows with secure role-based REST APIs and reviewer UIs, cutting manual operations by **100+ hours/month** and lifting throughput **60%**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ production RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core backend codebase from JavaScript to **TypeScript** across multiple repositories, tightening type safety and API contracts to make the production services more maintainable and easier to review.

Projects

Generative Models & Deep Learning From Scratch

github.com/Deva-1903/UMASS_CICS_689

JAX, PyTorch, NumPy | UMass CS 689 Advanced ML

- Implemented **RealNVP normalizing flows** and **DDPM diffusion models** from scratch in JAX, including custom forward/reverse processes, training loops, custom optimizers, and likelihood + sample-quality diagnostics on image benchmarks.
- Built and benchmarked **8 deep architectures** (CNN / ResNet variants, regularized models) on CIFAR-10; implemented **reverse-mode automatic differentiation over matrix-valued operations from scratch** to validate gradient correctness against framework autodiff, working through composition rules and graph traversal by hand.

AgenticSearch - Production-Grade Multi-Stage Service

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage Python/FastAPI service that converts open-ended queries into structured entity tables via typed retrieval planning, page rerank, deterministic parsers with LLM fallback, output verification, and cell-level provenance.
- Hardened the service with **150+ automated tests**, provider routing and fallback across upstream backends, pipeline instrumentation, and a reviewer-facing trust UI for inspecting extraction quality and ranking transparency, intentionally engineered for maintainability and reproducibility.

Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Databricks, SQL, Python

- Optimized a distributed Spark ETL pipeline on the NYC TLC taxi dataset by tuning broadcast joins, caching, partitioning, adaptive Spark settings, and column pruning to reduce shuffle volume and wall-clock runtime; profiled stage-level timings via Spark UI to localize shuffle and skew bottlenecks across naive vs. tuned runs.
- Documented a repeatable optimization workflow over the same dataset, capturing precomputed-field strategies and Spark configuration trade-offs so naive vs. tuned runs stay reviewable and reproducible for downstream analytics jobs.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.